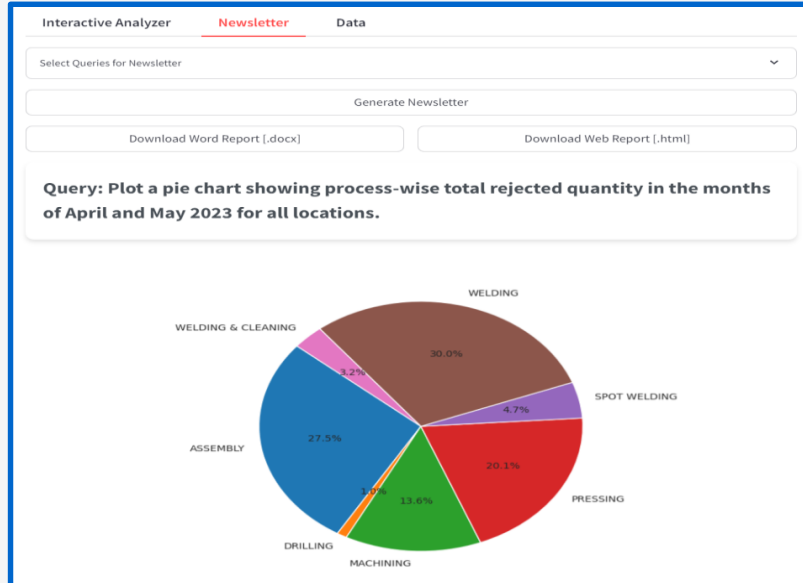




Enhancing Welding Work Through Data-Driven Solutions

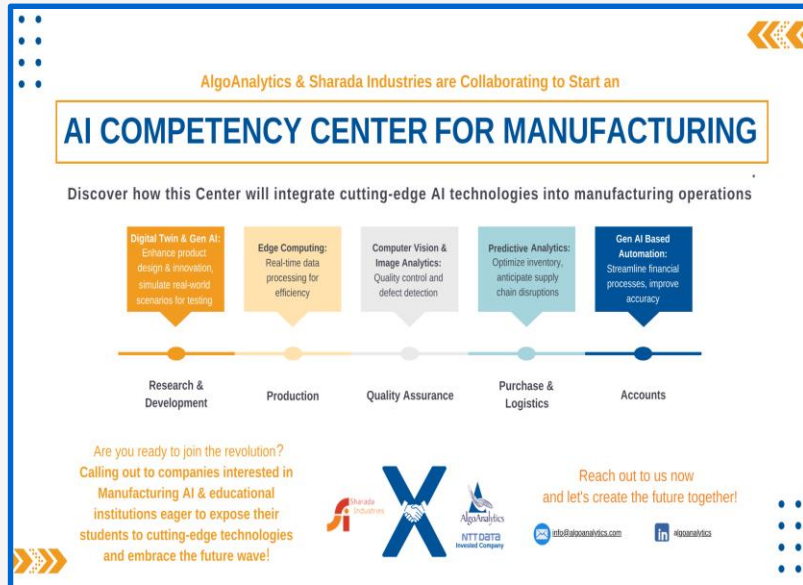
Advanced AI Solutions for Manufacturing Excellence

Overview of Welding Work



- Welding is a critical process in manufacturing, impacting product quality and durability.
- Data collection and analysis are essential for identifying defects and improving welding techniques.
- Utilizing advanced technologies like AI and sound analysis can enhance quality control.

AI-Driven Quality Control



- Implementation of sound analysis models to classify weld quality in real-time.
- Detection of welding defects such as lack of penetration, porosity, and spatter.
- Automated data collection from multiple modalities (audio, video, sensor data) ensures comprehensive monitoring.

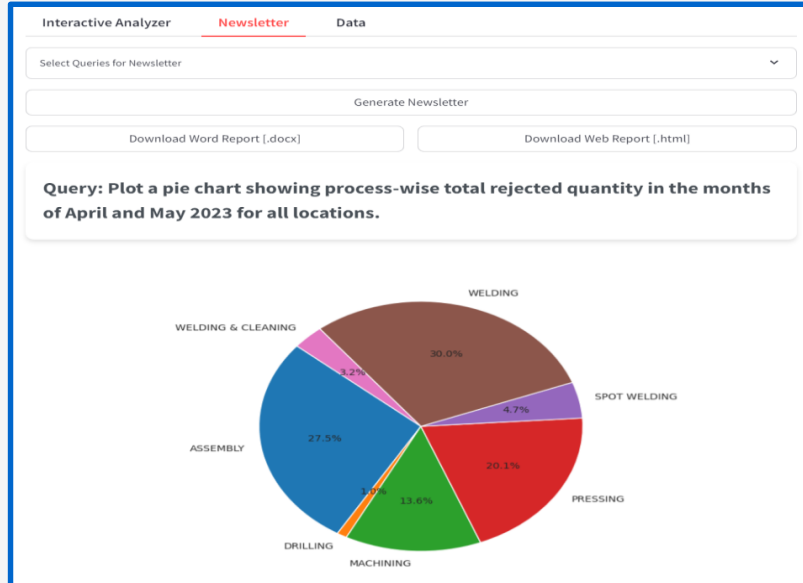
Case Study: Data Acquisition for Intel

Similar projects using image/video analytics:

- **Absenteeism Prediction** for workforce for a top global **automobile company**
- **Manufacturing process monitoring** for a large company in Ahmedabad, India

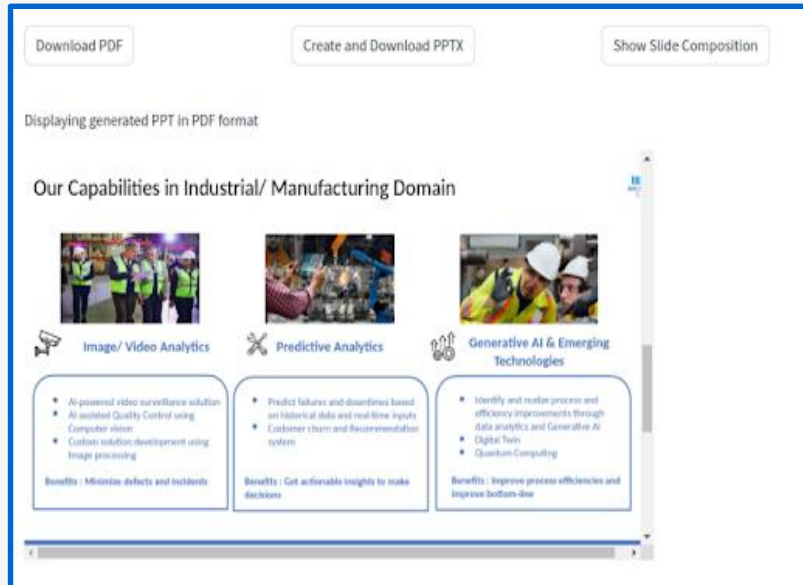
- Collaborative project with a global semiconductor manufacturer to collect extensive welding data.
- Setup included robotic welding machines equipped with cameras and microphones for synchronized data logging.
- Focus on identifying and documenting various welding defects to improve manufacturing processes.

Benefits of Enhanced Welding Data Collection



- Improved defect detection leads to higher product quality and reduced rework costs.
- Data-driven insights enable proactive adjustments in welding processes, enhancing efficiency.
- Comprehensive datasets facilitate better training of AI models for future applications.

Future Directions in Welding Technology



- Continued integration of AI and machine learning for predictive maintenance and quality assurance.
- Expansion of data collection techniques to include additional sensor modalities for richer datasets.
- Collaboration with industry leaders to refine welding processes and set new quality standards.